



BRAIN UP

A Book of
GENERAL KNOWLEDGE

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(M. A. English)

8

Hi! I'm



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Preface

The aim of the series is to connect our learners to the learning environment. An environment shaped and planned to run through the series.

The aim is to combine exiting, enjoyable moments of awareness with activities and stimulating research work that revalue around knowledge. The mission is to develop aptitudes in every child.

In short, '**Brain Up**' would encourage the readers to think, so that they become confident and exited about learning.

The series caters to grade levels 1 to 8, and brings the subject closer to life. It curiously connects to the multifaceted strums of knowledge, be it environment, science social studies, language and literature sports and so on... with the main aim, which is to boost a "love for learning and knowledge".

Key Features :

- An updated content or knowledge.
- Objective General Knowledge sections.
- Points to gain more knowledge.
- "Mind Booster" Contents
- Answers for reference.
- A Comprehensive and challenging range of knowledge— based worksheets, comfortable to and ideal for each grade level.



Contents

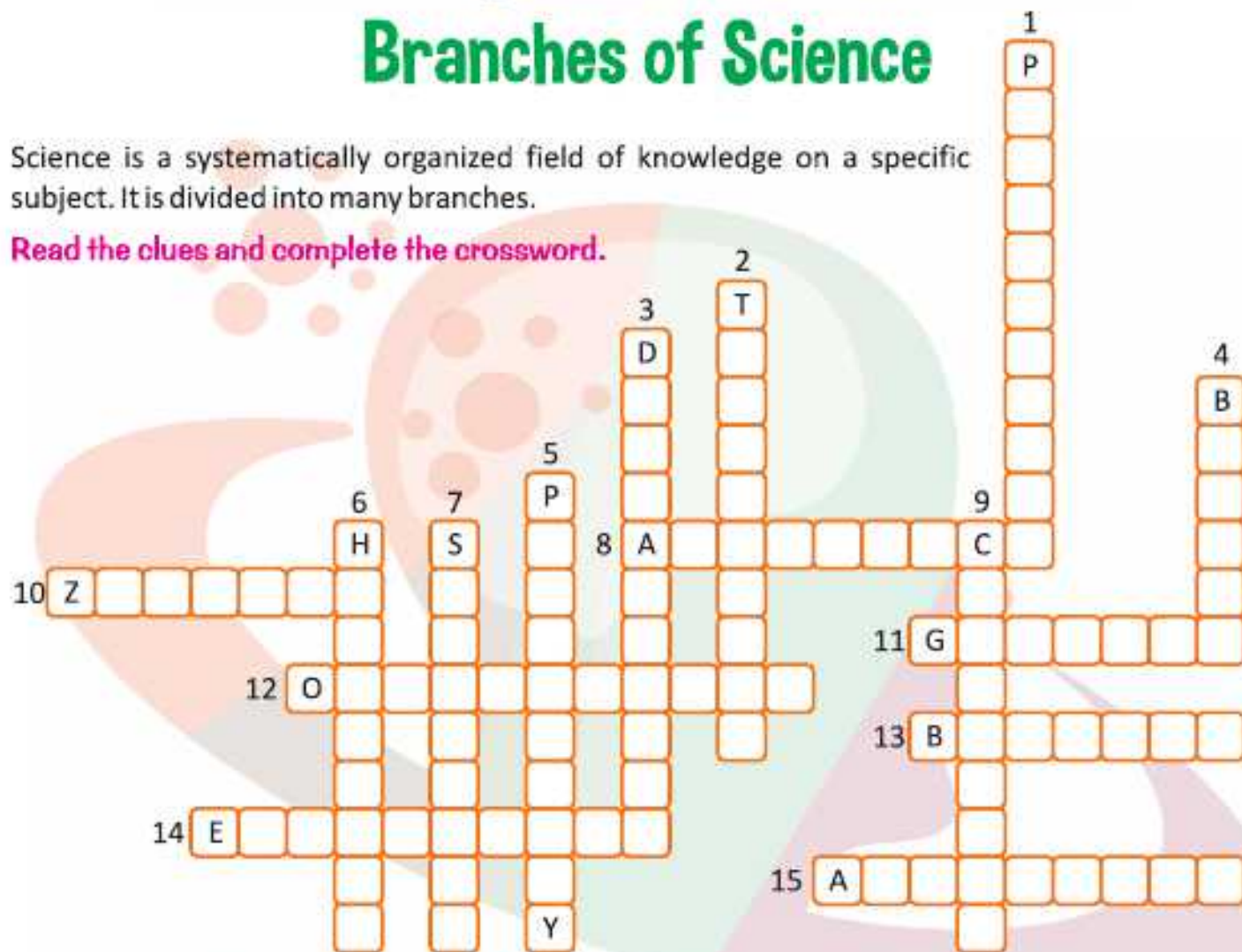
1. Branches of Science	5	25. Eminent Indians	41
2. Chandrayaan	6	26. Popular Cities	42
3. CN Tower	8	27. Woman Rulers in the History of India	44
4. Disguise to Hunt	10	28. Eastern Part of India	46
5. Aquatic Animals	11	29. Major Religions and Eminent Indians	48
6. Unique Animals	13	30. Memorials	50
7. Breeds of Dog	15	31. Famous Mathematicians	52
8. Birds of India	17	32. World Superlatives	54
9. Flying Animals	18	33. Ancient Greece	55
10. Pictures Making	19	34. Ancient Egypt (Pyramids)	56
11. From the Art Gallery	20	35. Tickle Your Brain	58
12. Computer Mania	22	36. Reasoning Test	59
13. Chemical Compounds	23	37. Odd One Out!	60
14. Astronomical Terms	24	38. Mind Test	61
15. The Aryabhatta Satellite	25	39. Letter Coding	62
16. Inventions File	26	40. Word Game	63
17. Nutritional Diseases	28	41. Direct Analogy	64
18. The Baobab Tree	30	42. All Occupations	66
19. Useful Plants	32	43. Proverbs	67
20. Water Facts	34	44. Idioms	68
21. Festivals of India	36	45. Literary Terms	69
22. Medieval India	37	46. Homophones	70
23. Special Names of Famous Indoor	39	47. Words Often Confused	71
24. River of India	40	Answer Sheet	72



Branches of Science

Science is a systematically organized field of knowledge on a specific subject. It is divided into many branches.

Read the clues and complete the crossword.



Across →

8. The study of sound.
10. The study of animals.
11. The study of the earth, its structure and history.
12. The study of birds.
13. The study of living things.
14. The study of insects.
15. The study of heavenly bodies in the universe.

Down →

1. The study of languages.
2. The study of poisons.
3. The study of skin and skin diseases.
4. The study of Plant life.
5. The study of human and animal behaviour.
6. The study of water.
7. The study of human society.
9. The study of the properties of matter.



Chandrayaan

Chandrayan-1 was the first Indian lunar probe under the Chandrayaan programme. it was launched by the Indian Space Research Organisation (ISRO) in October 2008, and operated until August 2009. The mission included an orbiter and an impactor. India launched in the spacecraft using a PSLV-XL rocket on 22 October 2008 at 00:52 UTC from Satish Dhawan Space Centre, at Sriharikota, Andhra Pradesh. The mission was a major boost to India's space programme.



Chandrayan-2, "moon craft" is the second lunar exploration mission developed by the Indian Space Research Organisation (ISRO), after Chandrayan-1 (2008). It consisted of a lunar orbiter, the Vikram Lander, and the Pragyan Lunar Rover, all of which were developed in India. The main scientific objective is to map and study the variations in lunar surface composition, as well as the location and abundance of lunar water.



Chandrayaan-3 is the third mission in the Chandryaan Programme, a series of lunar-exploration by the Indian Space Research Organisation (ISRO). The mission consists of a **lunar** lander named **Vikram** and **lunar rover** named **Pragyan**, similar to those launched aboard Chandrayaan-2 in 2019.



Key scientists and engineers involved in the development of Chandrayaan - 3

- Mohan Kumar — Mission Director
- Dr. P. Veramuthuvel — Project Director
- Kalpana Kalahasti — Associate Project Director
- K. Sivan — Former Chairman
- S. Somanath — Chair of Chandrayaan-3
- Kalpana Kalahasti — Deputy Project Director (Radio Frequency systems)

Tick (✓) the correct option.

1. According to ISRO, Chandrayaan - 1 moon mission will explore which area of moon?
(a) Moon's North Pole ☐ (b) Moon's South Pole ☐ (c) Moon's North-West part ☐
2. How many modules Chandrayaan - 2 space craft has ?
(a) 3 ☐ (b) 2 ☐ (c) 4 ☐
3. Chandrayaan - 2 Lunar mission Comprises of :
(a) An orbiter ☐ (b) Lander and Rover ☐ (c) All of these ☐
4. Which rocket carried Chandrayaan - 3 into the space?
(a) PSLV ☐ (b) GSLV ☐ (c) GSLV MK-III ☐
5. Lander of Chandrayaan-3 is also known as—
(a) Vikram ☐ (b) Vijay ☐ (c) Vikramaditya03. ☐





CN Tower

The world we inhabit is so rich and diverse. There is so much to see and experience. One gets to learn and be a richer and more knowledgeable person. Traveling to new places gives one the opportunity to interact with different people and experience new cultures and life-styles.

Among the numerous beautiful places in the world, one is Toronto, in Ontario, Canada. The famous majestic CN Tower, which remained the tallest free-standing structure for 34 years at a stretch leaves one spellbound. The CN Tower reigned supreme as the tallest tower in the world till 2010, when Burj Khalifa and Canton Tower crossed it in terms of height.



CN Tower stands for Canada's National Tower. It is a skyscraper 1,815 feet tall. The tower was constructed on the former railway land serves as an observatory and communication centre.

It is a popular tourist destination and enjoys the title of 'Face of Toronto's skyline'.

The plan to construct CN Tower was conceived in 1968 and construction started in 1973. The tower opened to the public on 26th June 1976. CN Tower is a symbol of Canada's global image. It is a radio and television communication centre, relaying more than 16 Canadian television and FM radio stations. More than 500 people are employed at the tower.



Tick (✓) the correct option.

1. CN Tower is known as:
(a) Canada's National Tower ☐
(b) Canada's New Tower ☐
(c) Canada's Natural Tower ☐
2. The restaurant gives:
(a) 90° view of Toronto ☐ (b) 180° view of Toronto ☐ (c) 360° view of Toronto ☐
3. CN Tower also offers adventure; the _____
(a) Edge walk ☐ (b) Cat walk ☐ (c) Bungee jumping ☐
4. The _____ of the tower attracts lightning!
(a) Shape ☐ (b) Height ☐ (c) Antenna ☐
5. The lifts in CN Tower take just _____ to reach the top of the tower.
(a) 58 minutes ☐ (b) 58 seconds ☐ (c) 48 seconds ☐



Disguise to Hunt

Do you know how these animals disguise to hunt their preys?

Identify the animals and write their names.

1. It lies on the bed of river and lakes with its mouth wide open. When a fish comes nearby, it shakes its tongue, which looks like a worm. Fish approaches to grab the worm and it closes its mouth.

2. It buries itself in mud at the bottom of the sea. When a small fish comes near, it emits a powerful electric charge to stun the fish. It makes easy to catch its prey.

3. It lurks in trees in southern Africa. Its bright green colour is identical to that of the leaves on the trees where it lives. This makes it almost impossible to see, unless it moves _____.
4. It lies on the bottom of the sea. It remains partially buried in sand. There is a flap of skin at the end of its spine. This flap looks like a worm. It waves it to attract small fish _____.
5. It feeds on ants and other small insects by setting a clever trap. The larvae digs a tunnel in sand and buries itself at the bottom. It appears to be a small hole to a passing ant. _____.





Aquatic Animals

Animals which live under water are known as 'Aquatic Animals'.

Read the following facts about aquatic animals.

1. Sea sponges have no head, mouth, eyes, feelers, bone, heart, lungs or brain, yet they also live.
2. A baby grey whale drinks enough milk to fill more than 2000 bottles a day.
3. Sea urchins have their mouth underneath their body.
4. Flying fish have an amazing pair of fins that fan out and look like wings.
5. At 39 cm in width, the Giant squid has the largest eye in the animal kingdom.
6. Walrus have balloons in their neck that fill up with air and keep their head above water while they sleep.
7. Starfish turns its stomach inside out and brings it out through its mouth.
8. Sharks seemingly are the only animals that never get sick, and are immune to every known disease including cancer.



Read the description and identify the fish given below.

1. They are found in tropical and temperate oceans around the world. They are large and powerful and grow up to 15 feet in length. This fish is named so because of its sword, which is a bony, sharp-tripped extension of the upper jaw. _____
2. They live in tropical Oceans. They are also called anemone fish because they live among sea anemones. Their unusual habitat and bright colour have made them a popular aquarium fish. _____
3. They are small-scaled with a single dorsal fin. Their sharp spines, which resemble a surgeon's spines, are located on either side of their tail base. These fish are mostly algal eaters. _____
4. They are named so for their bird-like beak, which they use to scrape up corals. They are tropical fish and have bright colours. They are found near coral reefs in the warm parts of the Atlantic, Indian and Pacific ocean. _____



Surgeonfish

Swordfish

Parrotfish

Clownfish

HELP BOX



Unique Animals

Everyone has some unique qualities.

Read about some unique animals.

1. **Addax** is a typical desert dweller with its large wide spreading hooves and feet adapted to walking on soft sand. It has a special ability to track rainfall and moves to the areas where desert vegetation will suddenly sprout after a downpour.



2. **Hamsters** are rodents that are often kept as a houses pets. It is originally from Syria. A hamster can store their food in their cheeks as they collect it and then take it to their colonies to eat it all later.

3. **Leafy Seadragon** has little leaf-like appendages which make it looks like a piece of seaweed. The leafy sea dragon can best manage to hide itself easily among the plants.



4. **Malabar Gliding Frog** that lives in the forests of the Western Ghats in India can fly. This tiny green frog glides by spreading the webbing between the finger and toes.

5. **Climbing Prech** is an amazing fish which can walk on land in search of water when its water hole dries up: They walk with jerky movements and supporting itself with the spiny edges of its gill plates and propelled by the fins and tail.



Fill the correct word with the help of option given in the bracket.

1. A species of _____ in Australia grows up to 10 feet length. (earthworm / snakes)
2. A Flamingo can only eat when it's _____ is upside down. (beak / head)
3. The strongest land mammals in the world is the _____. (Rhinoceros / African elephant)
4. _____ whales create the loudest sound among any living reature. (Blue / Humpback)
5. A tarantula spider can survive for more than _____ without food. (2 years / 2 months)
6. Owl, swallow their prey whole because they have _____ teeth. (no / two)
7. A woodpecker can peck upto _____ in a second (40 times / 20 times)
8. Penguins can convert saltwater into _____. (freshwater / limewater)
9. Koalas sleep an average of _____ a days. (20 hours / 22 hours)
10. A tiger's tail can be up to _____ of its total body length. (two-third / one-third)





Breeds of Dog

Dogs are our good friends and they are also very useful to us.

Let's read about some of the intelligent breed.

1. Golden Retriever is a large-sized breed of dog. It has an instinctive love of water, and are easy to train to basic or advanced obedience standards.
2. Papillon is a breed of dog of the spaniel type. It is also known as continental Toy spaniel. One of the oldest of the toy spaniels, it derives its name from its characteristic butterfly like look of the long and fringed hair on the ears.
3. German Shepherd are also called Alsatian. It is one of the most recognized breed of pets. They are very loyal dogs and are always aware of their surroundings. They have been used by police and military forces.
4. Boxer is a very energetic and playful dog. It is not aggressive but do need to be trained and socialised.
5. Havanese is small and quite lovable dog. It is extremely intelligent and is also recognized as Cuba's National dog.
6. Saluki is one of the most ancient dogs known to man , It clearly played a significant role in ancient Egypt. It is also known as 'Royal dog of Egypt'.



Identify the following breeds of dogs and write their names using the Help Box.



Bulldog Maltese Chihuahua Doberman Pug
Bullmastiff Dalmatian Poodle Blue Heeler

HELP BOX



DO YOU KNOW?

The Saint Bernard is a looking dog credited with saving the lives of some 2,500 people in 300 years of service as pathfinder and recess dog.



Birds of India

There are many common birds that you can see flying in the sky.

Read the information and write their name also.

1. This little bird can be seen riding on the backs of most cattle in the country side. It has yellow plumes on its head and neck during breeding seasons. _____
2. The most magnificent of all birds. It is no great flyer. It is admired for its spectacular plumage. Its tail feathers are covered by most Indians _____.
3. This bird is also seen pretty regularly in open marshy areas. The birds are always found in pairs and the male and the female generally stay together. _____.
4. It belongs to cuckoo family. It is known for its melodious songs. This bird is more often heard than seen. It never builds a nest but lays. It lays its eggs in a crow's nest. _____.
5. One of the most common birds you see fitting around in your garden. They often build its nest in the shrubs, in the homes and gardens. _____.





Flying Animals

We have seen flying birds, but there are also some animals who fly.

Name the following.

1. It is the fastest two legged animal on the plane, capable of running over 40 mph. It is a bird that can't fly _____.
2. Of all the gliding mammals, it has the most extensive adaptation to flight. In the air, it can glide as far as 70 metres from one tree to another. _____.
3. This flying squirrel is the most massive gliding animal known. It is found in the rugged, mountainous region of northern Pakistan. _____.
4. It is a flying lizard found in tropical rainforest in southern India and southeast Asia. It uses gliding only as a means of locomotion and not to escape from predators. _____.
5. It is the only mammal in the world, naturally capable of flight. Its most distinguishing feature is that its forelimbs are developed as wings help in flying. _____.



**DO YOU
KNOW?**

Wallace flying frog is a flying frog distinguish from other flying frog by its large size.



We all use different things to make pictures.

You all have used these things to make pictures. Can you answer these questions.

- Here are the three pictures of a dog. What has the artist used to make each one—crayons, a pencil or paints? Write under each picture.

(a)



(b)



(c)



- A picture which is made by pasting different things on paper is called a _____.
- Paints which you use by mixing with water are called _____ colours. Many grown up artists use paints which they mix with oil. Such paints are called _____.
- This mat has been made by _____ strips of paper. Did you know that cloth is made in the same way?
- These flowers have been made by _____ with a potato something with which you print is called a block.





From the Art Gallery

An art gallery is a room or a building in which visual art is displayed.

Let us read about some amazing forms of art.

1

*Calligraphy is
a kind of music
not for the ears
but for the eyes*
V. Lazursky

Calligraphy is the art of using a pen or brush to create fine hand-writing. It can be very decorative. It is the part of all civilizations, especially China and Japan.

2



Origami is the art of folding paper into three dimensional (3-D) shapes. In this form of art, the designer tries to make recognizable creatures such as birds, frogs and fish.

3



Mask Making is an art and craft associated with the theater, merry-making and some rituals where a person's identity may need to be hidden or altered.

4



Folk Art is performed by untrained artists following their local or inherited tradition of decoration and painting.

5



Mosaic is a decorative covering of wall, floor or similar surfaces with small pieces of stone, marble and glass. This technique is often used to decorate churches. It is very old technique.

6



Collage is a technique of sticking or gluing things like paper and cloth to a surface to make a work of art. This name of this form of art is derived from the French word coller, which means to glue or stick something.

Identify the different art forms and write their names by re-arranging the jumbled letters.

1



CSIMEACR _____

2



TIFGFARI _____

3



NITOAINMA _____

4



HYPAGROOTHP _____

5



REUPTLSUC _____

6



REUETCTHIACR _____



There are numerous short forms related to the world of computer,

Write down the full forms of the following.

1. COMPUTER → _____
2. CPU _____
3. PROM _____
4. HDD _____
5. SMPS _____
6. BIOS _____
7. VDU _____



Chemical Compounds

Chemistry is the main branch of science.

Identify the chemicals and write their common names in the space provided.

Chemical Name	Chemical Formula	Common Name
1. Ammonium Chloride	NH_4Cl	_____
2. Calcium sulfate Dehydrate	$\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$	_____
3. Acetic Acid	CH_3COOH	_____
4. Sodium Carbonate	Na_2CO_3	_____
5. Nitrous Oxide	N_2O	_____
6. Calcium Carbonate	CaCO_3	_____
7. Sucrose	$\text{C}_{12}\text{H}_{22}\text{O}_{11}$	_____
8. Trichloromethane	CHCl_3	_____
9. Calcium Hydroxide	Ca(OH)_2	_____
10. Sodium Hydroxide	NaOH	_____
11. Ethylene	C_2H_4	_____
12. Ethyl Alcohol	$\text{C}_2\text{H}_5\text{OH}$	_____
13. Silicon Dioxide	SiO_2	_____
14. Trinitrotoluene	$\text{C}_7\text{H}_5\text{N}_3\text{O}_6$	_____
15. Sodium Chloride	NaCl	_____

Chloroform	TNT	Chalk	Nasadar	Salt	Gypsum	Vinegar
Alcohol	Sugar	Sand	Laughing Gas	Caustic Soda	Washing Soda	
Ethene	Bleaching Powder					

HELP BOX



Astronomical Terms

Astronomy deals with celestial objects, space and the physical universe as a whole.

Write the correct answer of the following.

1. A device which can produce thrust by burning fuel and is capable of moving itself as a result of reaction _____.
2. A gigantic ball of ice and rock that orbit the sun in a highly eccentric orbit _____.
3. The study of celestial systems, including the solar system, stars, galaxies or any galactic clusters _____.
4. The original container for the astronauts in their orbiting flights around the earth _____.
5. A bowl-shaped depression formed by the impact of an asteroid or meteoroid _____.
6. A cloud of dust and gas in space, usually illuminated by one or more stars _____.
7. The path of a celestial body as it moves through space _____.
8. A natural or artificial body in orbit around a planet _____.



**DO YOU
KNOW?**

Space suit is a suit which can provide the necessary environment for an astronaut in case as cabin pressure failure.



The Aryabhata Satellite

Aryabhata was India's first satellite.

Find out the correct answer from the given words in the brackets.

1. The Indian satellite, Aryabhata was launched by the Soviet Union on _____.
(19 April 1975 / 19 August 1975)
2. It was built by _____ to gain expedient in building and operating a satellite in space.
(NASA / ISRO)
3. The Aryabhata satellite was fabricate at HAL _____. (Mumbai / Bangaluru)
4. In Aryabhata satellite, a total of _____ parameters were monitored. (19 / 91)
5. Body-mounted silicon solar panels and rechargeable NI-Cd chemical _____ were used in Aryabhata satellite. (Batteries / Compounds)
6. It carrier frequency for the _____ was 137.44 MHz. (Down link / Up link)
7. Its real time data transmission rate is _____ bites / sec. (265 / 256)
8. A PCM / FM / PM down link for _____ used in it. (Communication / Satellite)
9. Orbital period of Aryabhata satellite was _____ minutes. (96.46 / 94.64)
10. Aryabhata satellite has the capability to maintain the _____ temperature between 0 and 40 degree centigrade. (external / internal)
11. The Aryabhata satellite reentered the Earth's atmosphere on _____.
(11 February 1992 / 26 January 1992)
12. Aryabhata satellite was launched by _____ vehicle. (Kosmos-3M / Cosmos-2M)



**DO YOU
KNOW?**

Through Aryabhata satellite, first experiment was x-ray sources primarily in relation to their time variation effect in energy range of 2.5-150 kev.



Inventions File

Inventions are the main aspect of science.

Read about the following inventions.

1. The Microwave Oven was invented by an American engineer and inventor Percy Lebaron spencer in 1947. In 1945, he created a device to cook food using microwave radiation. The first microwave Oven was called the Retiring. This common kitchen appliance was discovered by an accident.
2. The radio was invented by an Italian by an Italian inventor, Guglielmo Marconi. He was born on 25 April 1874. He sent and received his first radio signal in Italy in 1895.
3. The microphone is an instrument that converts sound waves into electronic signals. It is invented by an American inventor Thomas Alva Edison.
4. The digital clock was first invented by Joses Pallweber in 1956 using a collaboration methods and techniques. It is a type of clock that displays the time digitally.
5. The first successful airplane was invented in 1903 by Orville and Willbur wright. They were two Americans who also known as wright brother. They also developed a heavier than airplane that moved through the air by the power of its own motor.



Find out the Inventor of these famous inventions.

	Invention and Year	Inventor
01.	Artificial Heart (1961)	_____
02.	Atomic Bomb (1945)	_____
03.	Ballpoint Pen (1888)	_____
04.	Diesel Engine (1892)	_____
05.	Electric Iron (1882)	_____
06.	Piloted Helicopter (1907)	_____
07.	Dynamic Loudspeaker (1898)	_____
08.	Pencil (1795)	_____
09.	Printing Press (1450)	_____
10.	Safety Razor (1901)	_____
11.	Steam Engine (1698)	_____
12.	Motorized Vacuum Cleaner (1901)	_____
13.	Paper clip (1899)	_____
14.	Video Games (1972)	_____
15.	Bifocal Lenses (1784)	_____
16.	Seaplane (1910)	_____
17.	Machine Gun (1884)	_____
18.	Electric Hair Dryer (1890)	_____
19.	Contact Lens (1888)	_____
20.	Detergent (1933)	_____



The first known artificial refrigerator was demonstrated by William Cullen at the University of Glasgow in 1748. Before the invention of the refrigerator, people used to preserve their food by placing them near freshwater lakes or packed with snow and during the winters.



Nutritional Diseases

A prolonged deficiency of nutrients can result in some serious diseases, such diseases are called nutritional or deficiency diseases. Diseases caused by the deficiency of Protein.

1. Kwashiorkor is caused by the deficiency of protein and carbohydrate intake. It most often afflicts children in poor countries where food supply is limited.



2. Marasmus is also a form of malnutrition. It generally afflicts in infants and causes severe loss of fat, muscles and other tissues of the body. This Disease is caused by the deficiency of Vitamins.

3. Night Blindness is not a complete lack of vision at night as the name implies. A person suffering from this disease cannot see well at night or in poor light.



4. Beriberi are of two types. One type affects the heart and the other damages nerves and muscles. It is caused by deficiency of Vitamin B in diet.

5. Scurvy is a disease that affects blood vessels, skin and the healing process of the body. It results in the formation of spots on the skin and swollen gums which often bleed. It is caused by the deficiency of Vitamin C.



6. Rickets causes a child's bone to become soft and weak. Because of this, the bones become prone to fractures and deformity. The main cause of rickets is the deficiency of Vitamin D in the diet.

Match the following disease causes due to the lack of which nutrition.

Disease	Nutrition
1. Osteoporosis	(a) Protein and Carbohydrate
2. Beriberi	(b) Iron
3. Scurvy	(c) Vitamin B
4. Kwashiorkor	(d) Calcium
5. Anaemia	(e) Vitamin C





The Baobab Tree

Ever heard or seen a giant tree, a tree whose trunk you can enter? One such tree is the Baobab tree. There are nine species of the Baobab tree (genus *Adansonia*); six are found in Madagascar, two are in Africa and one in Australia. These are all commonly known as Baobab, a word derived from the Arabic word "buhibab", meaning a fruit with many seeds. The Baobab tree has the distinct feature of swollen stems.

The Baobab is a deciduous tree, having a height of 5 to 20 metres. It grows to a mammoth size and according to carbon dating it lives up to 3,000 years. The Baobab tree is originally from the dry and barren African Savannah. It symbolizes life and positively in an otherwise barren land. The Baobab tree has adapted to its environment and is quite luscious, indicating that during the monsoons, it absorbs water and hoards in its mammoth trunk. No wonder, then, it produces a fruit rich in nutrient in a scorched land. The Baobab tree has the capacity to store nearly 31,700 gallons (120,000 litres) of water that helps it survive the harsh drought conditions. Thus, it has come to be known as "The Tree of Life".



Some of us would have certainly heard of the Baobab tree. But that it is a fruit bearing tree and that too one of the most nutritional fruit in the world is a lesser known fact. The fruit grows up to a foot long and has tartaric acid and vitamin C. It can be sucked or soaked in water with the aim of preparing a refreshing drink.

Tick (✓) the correct option.

1. There are _____ species of the Baobab tree.
(a) Twenty (b) Nine (c) Three
2. The Baobab tree is a _____ tree.
(a) Flower (b) Leafy (c) Fruit bearing
3. There are only two species of Baobab trees in _____.
(a) India (b) Africa (c) Australia
4. The Baobab tree has the distinct feature of _____.
(a) Swollen stems (b) Massive roots (c) Yellow leaves
5. The Baobab is a _____ tree.
(a) Deciduous (b) Pecan (c) Oak





Useful Plants



There are some variety of plants which are useful to us.

Let's read out some facts about plants.

- When the fruit of Hemlock is eaten, it induces paralysis. It was said that this plant was used to kill the great Greek Philosopher, Socrates.
- The leaves of Mimosa (touch-me-not) go limp and droop when touched by someone.
- Tulsi, an aromatic herb, has medicinal value.
- The oil from tulsi leaves is used to destroy bacteria and insects.
- Bonsai is the art of developing miniature potted trees. These trees are dwarfed by stem and root pruning and controlled nutrition.
- In the 15th century, women would drop extract of deadly nightshade (belladonna) in their eyes to make them appear attractive.



Given below are examples of how some plants adopt to survive. Read the clues and re-arrange these letters.

1. It is an aquatic plant and it needs to keep itself sinking. Thus, its leaves are filled with air and its floats. (RTEWA IHTYCHAN) _____
2. This parasitic plant lacks trees roots but has root like structures buried in the branches of a tree. It depends upon the tree to anchor itself and absorb water and nutrients from host. (EEOTSILTM) _____
3. These trees grow in the tropical deltas and along ocean edges and river estuaries. They have wide-spreading stilt roots to stay erect in the tidal mud and trap nutritious organic matter. (SEORVNMAG) _____
4. This plant gets its name from the fact that it literally comes back from the brink of drying up. It can survive long periods of heavy drought by curling up and appearing dead. As soon as it gets a little water, it uncurls and reopens. (NOTCIERURSRE NEFR) _____
5. This desert plant is known for the capacity of its tissues to retain water. During the heat of the day, its stomata remain closed and do not allow the loss of moisture. (OARUAGS) _____





Water Facts

Water is a main source for living beings.

Let's read about it.

1. 60% of your body is made up of water.
2. You can survive weeks without food, but no more than a few days without water.
3. You gain water by drinking and eating, as a by - product of cell activity.
4. You lose water by sweating and breathing in your nose and skin.
5. The average person loses 1.5 liters of water everyday in urine, 0.5 liters in sweat, 0.3 liters as vapour in the breath, and 0.2 liters in feces.
6. The water balance in the body is controlled mainly by the kidneys and adrenal glands.
7. The amount of water the kidneys let out as urine depends on the amount of salt there is in the blood.
8. If you drink a lot, the saltiness of the blood is diluted (water down). To restore the balance, the kidneys let out a lot of water in the form of urine.



9. If you sweat a lot during heavy exercise you need to make up for all the water you have lost by drinking. Your kidneys make sure that if you chink too much you lose water as wine.
10. It you drink little or sweat a lot, the blood becomes more saltily. So the kidneys restore the balance by holding on the more water.

Tick (✓) the correct option.

1. The changing of water into vapour is—
(a) Evaporation ☐ (b) Sublimation ☐ (c) Relacation ☐
2. Excessive rain causes—
(a) Drought ☐ (b) Clouds ☐ (c) Floods ☐
3. The main source of water is—
(a) Tubewell ☐ (b) Rain ☐ (c) Handpump ☐
4. The solid form of water is—
(a) Steam ☐ (b) Ice ☐ (c) Smoke ☐
5. The circulation of water between ocean and land is—
(a) Water cycle ☐ (b) Carbon cycle ☐ (c) Nitrogen cycle ☐





Festivals of India

Different types of festivals are celebrated in Indian states.

Given below are a few festivals celebrated in India. You have to tell the state where they are mainly celebrated.

Festivals		State
1 Kavadi Attam		
2 Carnaval		
3 Shigmo		
4 Pang Lhabsol		
5 Losar		
6 Gangaur		
7 Uttarayani Mela		
8 Dosmoche		
9 Bhogali Bihu		
10 Chhath Puja		
11 Ugadi		
12 Ratha Yatra		
13 Baisakhi		



Medieval India

Medieval India refers to a long period of the history of the India subcontinent between the "ancient period" and "modern period".

Tick (✓) the correct option.

1. What was the main responsibility of Mir Bakshi during the Mughal Era?
(a) Food Department ☐ (b) Military Department ☐
2. Which battle helped Sher to establish Suri Dynasty?
(a) Battle of Chausa, 1539 ☐ (b) Battle of Haldighati, 1576 ☐
3. In the India history, which period is called as the "Golden Age of Architecture"?
(a) Gupta Period ☐ (b) Mughal Period ☐
4. Who forbade the inscription of Kalma on coins?
(a) Aurangzeb ☐ (b) Badauni ☐
5. Where is the famous Virupaksha temple located?
(a) Vijayanagar ☐ (b) Hampi ☐
6. Who was the ruler of the Vijaynagar Empire, when Raja Wodeyar founded the kingdom of Mysore?
(a) Krishnadevaraya ☐ (b) Venkata II ☐
7. What was the unit of measurement of land in Martha dominions?
(a) Kathi ☐ (b) Jarib ☐

8. Which scholar in the Mughal court alleged that Akbar was an enemy of Islam?
(a) Badauni (b) Bahur
9. In the year, 1613, where was the English East India company given permission to set up a factory.
(a) Agra (b) Swet
10. What was the single biggest item of import to the Vijaynagar Empire?
(a) Dogs (b) Horses
11. Which place in Bihar is related to birth of Sher Shah Suri?
(a) Sasaram (b) Saharsa



Special Names of Famous Indoor

Some people are known to us by special names.

Write the special names often tagged to these famous Indians.



Tuti-yi-Hind	Birdman of India	Indian Machiavelli	Nightingale of India	Netaji
Swar Kokila	Missile Man	Haryana Hurricane	Flying Sikh	

HELP BOX



River of India

The rivers of India play an important role in the lives of the people.

Read the clues and identify them.

1. This is one of the largest rivers of India. It originates in the Himalaya as the Tsangpo. It flows through Tibet and enters into India. _____
2. The Gangotri Glacier is the source of this river. It starts at Gaumukh as Bhagirathi. It joins Alaknanda at Devprayag. It flows through such towns of religious importance as Haridwar, Varansi and Allahabad before it enters the Bay of Bengal. _____
3. This is another holy river of North India. It is said to be the sister of the Hindu God of Death, Yama and the daughter of Surya. _____
4. It flows through the Deccan plateau. The source of the river is Talakaveri, a famous pilgrimage and tourist spot of Karnataka. The cities of Bangalore and Mysore depend on this river for their drinking water supply. _____
5. The river, originates at Mahabaleshwar in Maharashtra passes through Sangli and meets the sea in the Bay of Bengal at Hamsala Deevi in Andhra Pradesh. It flows through the states of Maharashtra, Karnataka and Andhra Pradesh. _____

Yamuna River Krishna River Brahmaputra River Ganga River Kaveri River

HELP BOX



Eminent Indians

Eminent Indians surveys the lives and contributions of some of the saints and sages, artists, musicians, dancers, film personalities, economists, litterateurs, political thinkers and legal luminaries of India.

Read about the great souls of India and try to identify them.

1. He was the teacher of first Mauryan Emperor Chandragupta. He is also known as the Kautilya and Vishnugupt. He is considered as the pioneers in the field of economics and political science. He wrote a book named 'Arthashastra'.
2. He was one of the principal science of Ayurveda. He is sometimes referred to as the Father of Indian Medicine.
3. He was an Indian mathematician and astronomer. He has been called the greatest mathematician of medieval India. His main work 'Siddharanata Shiromani' is divided into four parts & Lilavati, Bijaganita, Charaganita and Goladhyaya.
4. He was a Sanskrit grammarian. He is known for his formulation of the 3,959 rules of Sanskrit grammar which is known as 'Ashtadhyayi'.
5. He was an ancient Indian surgeon and is the author of the book 'Susruta Samhita'. This book he described over 300 surgical procedures and 120 surgical instruments. He is also known as the 'Father of Surgery'.



Panini

Chanakya

Sushruta

Charak

Bhaskaracharya

HELP BOX



Popular Cities

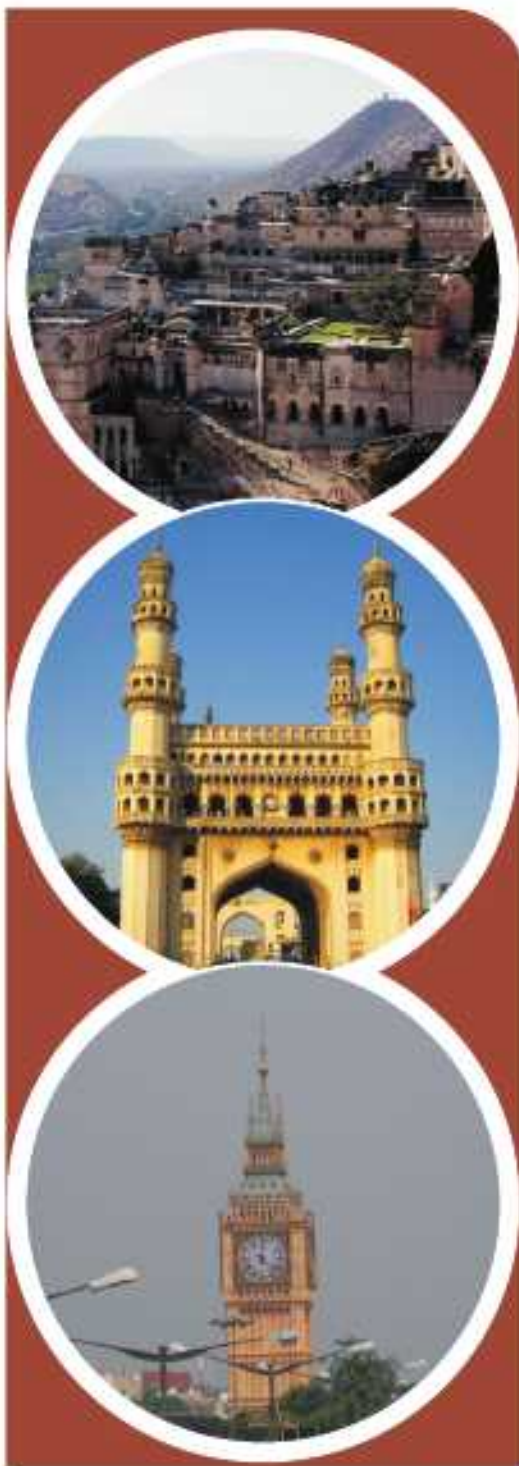
Cities of India are famous in all over the world.

Read the description of some popular cities of India, and write the names of these cities.

1. It is considered as the economical Capital of India. It is also the main centre of many famous Hollywood movies. It is the Capital city of the Maharashtra.

2. It is one of the India's most important and largest information technology hub. This beautiful city is known as the 'Garden city of India' _____.
3. It is the Capital city of Rajasthan. It is one of the major business hub for diamond, gold and the most important stone jewellery. Popularly this city is known as 'Pink city' _____.
4. It is located in Jharkhand. It is known as the 'Coal Capital of India' and it famous for its coal mining. The Indian school of Mines, is also situated here
_____.
5. It is popularly known as 'The City of Nawabs'. The city is on the northwestern shore of Gomti River, which flows through it. It is the capital city of Uttar Pradesh. It was founded by Nawab Asaf-ud-Daula.





6. It is surrounded by the Aravalli Mountains. It is a popular Muslim pilgrimage centre of the shrine or Dargah of sufi saint Khwaja Moinuddin Chishti. It was once known as Ajaymerree, the city once ruled by Prithviraj Chauhan _____.
7. It is the capital city of Andhra Pradesh. Popularly it is known as the name of 'The City of Pearls'. It is home for large number as it companies. The other industries which have boomed in and around it are of pharmaceutical and biotechnology _____.
8. It is placed on the banks of Hooghly River. It is known for its deep rooted culture and rich historical backdrops. Due to its historical past, it is awarded as the 'cultural capital of India' and the 'City of Joy'.



Lucknow
Hydrabad

Kolkata
Bengaluru

Ajmer
Dhanbad

Mumbai
Jaipur

HELP BOX





Woman Rulers in the History of India

There are few women who ruled during Medieval India.

Read the clues and identify them.

1

She belonged to the Kakatiya dynasty on the Deccan Plateau. She was the daughter of King Ganapathideva. She succeeded her father when she was only 14 year old _____.



2

She ruled over Gondwana from 1548 on behalf of her son Bir Narayan after the death of Dalpat Shah the ruler of Gondwana _____.



3

She is also known as Chand Khatun or Chand Sultan. She was the daughter of Hussain Nizam Shah 1 of Ahmednagar _____.



4

She ruled over Ahmednagar from 1766 to 1795. She was the daughter of Monakoji Shinde _____.



5

She married to Raja Mullasarja of Keituri, a princely state of Belgaum in Karnataka. Her husband died in 1816 leaving behind a son, who passed away in 1824 _____.



6

She was born in Varanasi and named Manu. She was married to Gangadhar, the ruler of Jhansi in 1842 _____.



7

She was married to Vikramaditya Singh, the ruler of Ramgarh state. Vikramaditya Singh died leaving behind his wife and no heir to the throne _____.





Eastern Part of India

Eastern part of India is mainly consisting of the India states of Bihar, Jharkhand, West Bengal and Odisha and also the union territory-Andaman and Nicobar Island.

A. Given below are the pictures of some of the places of interest of East India. Identify them and write their names.

1



West Bengal

2



Jharkhand

3



Bihar

4



West Bengal

5



Odisha

6



Odisha

7



**Andaman and Nicobar
Island**

8



West Bengal

B. Tick (✓) the correct option.

1. Which of these styles of painting originated in Bihar?
 (a) Warli ☐ (b) Madhubani ☐ (c) Phad ☐
2. Which of these ancient kingdoms existed in the eastern part of India?
 (a) Koshala ☐ (b) Panchala ☐ (c) Magadha ☐
3. In which of these states is the Bokaro Steel Plant located?
 (a) West Bengal ☐ (b) Odisha ☐ (c) Jharkhand ☐
4. Who among these was an exponent of Odissi dance?
 (a) Kelucharan ☐ (b) Uday Shankar ☐ (c) Rukmini Devi Arundale ☐
5. Which of these crops is abundantly grown in eastern India?
 (a) Jute ☐ (b) Jowar ☐ (c) Sugarcane ☐
6. In which state is Belur Math, the headquarters of Ramakrishna Mission, situated?
 (a) Odisha ☐ (b) Jharkhand ☐ (c) West Bengal ☐
7. Which lake, found in eastern India, is the largest coastal lake in India and one of the largest in the world?
 (a) Rabindra Sarobar ☐ (b) Chilika Lake ☐ (c) Nal Sarovar ☐

National Library of India

Darjeeling Himalayan Railway

Jagannath Temple

Victoria Memorial

Tagore Hill

Bodh Gaya Temple

Konark Sun Temple

Cellular Jail

HELP BOX



Major Religions and Eminent Indians

Religion is a system that relates humanity to spirituality. People around the world follow different religions. Our country is also the birth place of major religions.

Let us read about them.

1. The Hindus worship in temple. Though they believe in one God, yet they also believe in different forms of God such as Brahma Vishnu and Mahesh. Hindu pilgrims visit the holy city of Varanasi in a large number.
2. Muslims also believe in one God. They call it Allah. Their religion is called Islam. Muslims go on pilgrimage to Mecca. They pray in Mosque.
3. Christians believe in one God. They worship in Churches. The Vatican city is a religious centre for the Christians.
4. Buddhists follow the teachings of Gautam Buddha. One can see statues of Gautam Buddha in the Buddhist temple, which are popularly known as the Pagodas.
5. In Punjabi language the word Sikh stands for a



learner. Thus the most important thing for a sikh is to learn about God. Sikhs worship in the Gurudwaras.

6. The name Jain is taken from the title Jina, which means the one who wins himself. Jains respect 24 Jins, whom they call the Tirthankaras.



Tick (✓) the correct option.

1. Who is Wahiguru in sikh religion? (God / Guru)
2. Who was the first sikh Guru? (Guru Nanak Dev / Guru Granth Sahib)
3. Who was the last Tirthankara of Jainism? (Mahavira / Parshvanath)
4. The celebration of Christ's victory over death is celebrated as? (Good Friday / Easter)
5. What event in the Islamic history is commemorated in the fast during the month of Ramzan? (Muhammad's birth / Muhammad's first revelation from God)





Few memorials, which are constructed to commemorate some great people of our country.

Read these clues and identify these memorials.

1. It is a memorial of Mahatma Gandhi. Originally, it was the name of a historic ghat of old Delhi on the bank of River Yamuna. It is a black marble platform that marks the spot of Mahatma Gandhi's cremation on 31 January 1948, a day after his assassination _____.



2. It literally means Garden of Peace. It is the memorial of our first prime minister Pt. Jawaharlal Nehru. He was born on 14 November 1889 and passed away on 27 May 1964. It is located just adjacent to Raj Ghat and is a very peaceful place _____.

3. It is also known as the victory Bank. It is the memorial of Lal Bahadur Shastri. He was the second Prime Minister 1966. This memorial is also situated on the bank of River Yamuna _____.



4. It is the memorial of the former Prime Minister Rajive Gandhi. The design of the memorial consists of two squares in angular relationship signifying the day of his birth and the day of his assassination. The center of the square is a stone platform with lotus flowers carved in granite stone _____.



5. Dr. Shankar Dayal Sharma was an Indian scholar and politician. He is most notable for serving as the President of India from 1992 to 1997. He was cremated near the Vijay Ghat. Literal meaning of his memorial is Land of Duty _____.



Karma Bhumi

Veer Bhumi

Shanti Vana

Raj Ghat

Vijay Ghat

HELP BOX



Famous Mathematicians

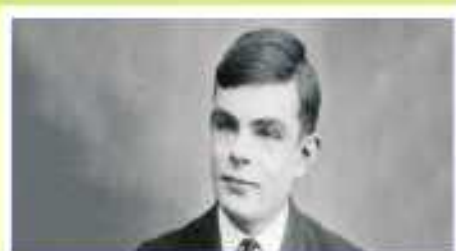
There are many mathematicians across the world, who have gained a lot of fame in their work.

Let us read about some of them.

1. Abraham de Moivre pioneered the development of analytic geometry and the theory of probability. He is known for his formula, $(\cos x + i \sin x)^n$, which took trigonometry into analysis. He found that he was sleeping 15 minutes longer each night and from this the arithmetic progression he calculated that he would die on the day that he slept for 24 hours. He was right.
2. Charles Lutwidge Dodgson was a famous British mathematician who worked mainly in the field of logic. He is much better known by his pen name. Lewis Carroll, that he used to write children's book, of which many included logic puzzles. He is also known for adding words to the English such as 'Chortle', which come from the poem 'Jabberwocky'.
3. Pythagoras was a famous Greek mathematician. He was the founder of the Pythagorean cult which was considered by Aristotle as one of the first groups that have extensively studies advance mathematics. He was known for his Pythagorean theorem in trigonometry. His theories saved for the development of geometry which labeled him as the 'Father of Modern Mathematics'.
4. Euclid was a renowned Mathematician who was famously called as the 'Father of Geometry'. He discovered the subject and gave it its value, making it one of the most complex forms of mathematics at the time. He began developing his theorems and collated it into a colossal treatise called 'The Elements'. He is also known for his logical proofs to several contentions and theorems.
5. Alan Turing was a British mathematician and computer scientist who cracked German Ciphers in the II world war. He contributed to mathematical logic and played an important role in development of algorithms, artificial intelligence and the modern computer.



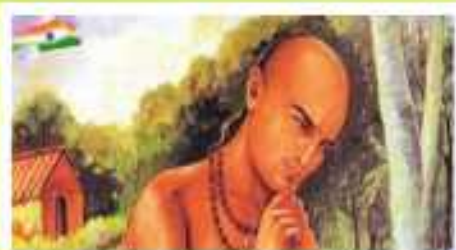
Identify these famous Indian mathematicians using help box.



1. He was born in 476 A.D. in Kerala
He gave the formula $(a+b)^2 = a^2 + b^2 + 2ab$

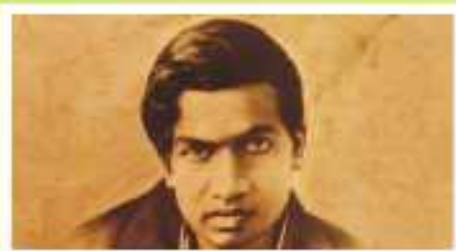
2. He was born in a village of Mysore district

He gave the formula like $\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$.



3. He was born on 22 December 1887 in Madras.

He showed that how to divide the number into two or more squares and cubes.



4. He was born in 598 A.D. in Bhinmal, Rajasthan.

He gave the formula for the area of a cyclic quadrilateral as:

$$K = \sqrt{(s-a)(s-b)(s-c)(s-d)}$$



Bhaskaracharya

Srinivasa Ramanujan

Brahmagupta

Aryabhatta

HELP BOX



World Superlatives

World is full of natural magnificent superlatives.

Answer the following.

1. This lake in the south central part of Siberia is 5,712 feet (1.7 km) deep. It is about 25 million years old and contains 20 percent of the Earth's fresh liquid water. What is the name of this lake? _____

2. It is a plateau in South America. This is the desert of all place known to man. No one remembers when it rained here last, which is the driest desert in the world? _____

3. It is the world's tallest statue with a height of 182 metres (597 feet)

located near Kevadia in the state of Gujarat, India. It depicts Indian statesman and independence activities. Sardar Vallabhbhai Patel, the first Deputy Prime Minister. What is the name of this statue? _____

4. The deepest depression in the oceans is the Mariana Trench more than seven miles down. It is the deepest point of the Earth. In which ocean is it located? _____

5. A cave is an opening into the Earth. The longest cave in the world has 405 miles of surveyed passage ways. Its name is suggestive of its size. Which is the longest cave of the world?





Ancient Greece

Ancient Greece was one of the greatest civilization in history. They put an emphasis on the value of the person and education.

Identify and write the names of them in the space provided.

1. He was a great philosopher of Greek. He was the student of Socrates. He wrote many dialogues using Socrates as a major character. He also founded the Academy in Athens.

2. He was a great astronomer and mathematician. He was the first to put the sun at the centre of the known universe rather than the Earth. _____
3. He was a great historian of Greek, who was known for the exact science of his research. He wrote about the river between Athens and Sparta. _____
4. He is considered one of the great mathematicians and scientists in history. He made many discoveries both in Maths and Physics including many inventions. _____
5. He was probably the most popular playwright during Greek times. He won many writing competitions, and is though to have written over 100 plays. _____



**DO YOU
KNOW ?**

The book that was about Greek Rural life called 'Works and Days' was written by Hesiod.



Ancient Egypt [Pyramids]

Pyramids are the large structure built especially in ancient Egypt.

Let us read about them!

1. **Why were the Pyramids built?**

The pyramids were built as huge monumental tombs for Egypt's pharaohs (kings). The Egyptians believed that dead people's spirits could live on after death if their bodies were carefully preserved. It was specially important to preserve the bodies of the pharaohs and nobles. Their spirits to survive. So the Egyptians preserved their dead bodies as mummies, and placed them in these splendid tombs along with cloths, jewels and models of everything they would need in life after death.

2. **Where are the Pyramids?**



In Egypt, in North Africa. They stand up on the west bank of the river Nile. The Egyptians believed this was the land of the dead, because the sun set there.

3. **How old are the Pyramids?**

The first step pyramid was built between 2630 and 2611 BC. Before them, people were buried under flat-topped mounds called "Mastabas", and in pyramids with stepped sides. The last pyramid was built about 1530 BC.

4. Why was the river Nile so important?

Egypt received hardly any rain, but every year the river Nile flooded, covering the fields along its banks with fresh water and fertile black soil excellent for growing crops. Farmers dug irrigation channels to carry water to more distant fields.

5. How were corpses mummified?

Making a mummy was a complicated and expensive process. First, a dead person's soft internal organs were removed, then the body was packed with chemicals and left to dry out. Finally, it was wrapped in resin soaked linen bandages and placed in a coffin.

6. How many kinds of boat were used in the river Nile?

Many kinds! Rafts made from papyrus reed, flat bottomed punts, big, heavy cargo boats splendid royal barges and funeral boats carrying bodies across the river to the pyramid tombs.

7. What are the pyramids made of?

Oh heard, smooth limestone, good quality stone was used for outer casing and poor quality stone and rubble for the inner core.



Tick (✓) the correct option.

1. What was the main purpose of the Ancient Egyptian pyramids?

(a) As temples to their Gods.

(b) As burial sites for the Pharaohs.

(c) As something to do to keep their slaves busy.

☐☐☐

2. Which Pharaoh built the first pyramid?

(a) Djoser

☐

(b) Khufu

☐

(c) Menus

☐

3. What is the largest pyramid built by the Egyptians?

(a) The Pyramid of Djoser

☐

(b) The Pyramid of Khafre

☐

(c) The Great Pyramid of Giza

☐

4. What geometrical shape formed the base of all the pyramids?

(a) The rectangles

☐

(b) The rhombers

☐

(c) The square

☐

5. About how tall is the pyramid of Khufu at Giza?

(a) 250 feet

☐

(b) 480 feet

☐

(c) 120 feet

☐



Tickle Your Brain

It is said that a hearty laugh and a good night sleep can solve most of your problems. Laughter is great for your physical and mental well-being.

Answer the following questions based on arithmetic operation.

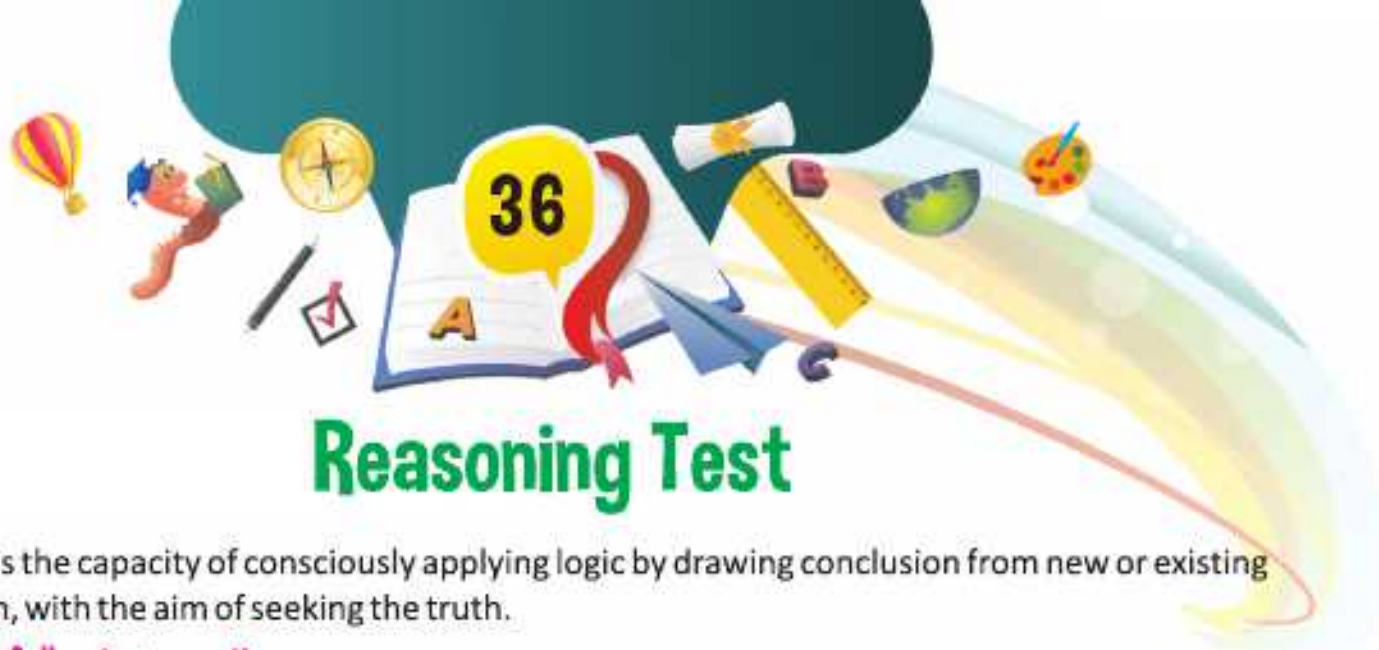
1. In a class of 60, where boys are twice that of girls. Tanya ranked 17th from the top. If there are 9 boys ahead of Tanya, how many girls are after her in the rank?

2. Rahul is 7th from the left and Riya is 5th from the right. When they interchange their position, Riya becomes 19th from the right. What is Rahul's position from the left?

3. The ratio of ducks and frogs in a pond is 37:39 respectively. The average number of ducks and frogs in the pond is 152. What is the number of frogs in the pond?

4. When 3626 is divided by the square of a number and the answer so obtained is multiplied by 32, the final answer obtained is 2368. What is the number?

5. Flight to Mumbai leaves every 5 hours. At the information counter, I learnt that the flight took off 25 minutes before. If the time now is 10:45 am, what is the time for the next flight?



Reasoning Test

Reasoning is the capacity of consciously applying logic by drawing conclusion from new or existing information, with the aim of seeking the truth.

Answer the following question.

1. What numbers are missing in the following?

1	3	8	19	42	?	Ans.	
3	7	16	35	74	?		
2	5	12	27	58	?		

2. Name the smallest natural number the spelling of which contains the letter 'a', 'b', or 'c'?

3. What is the missing number?



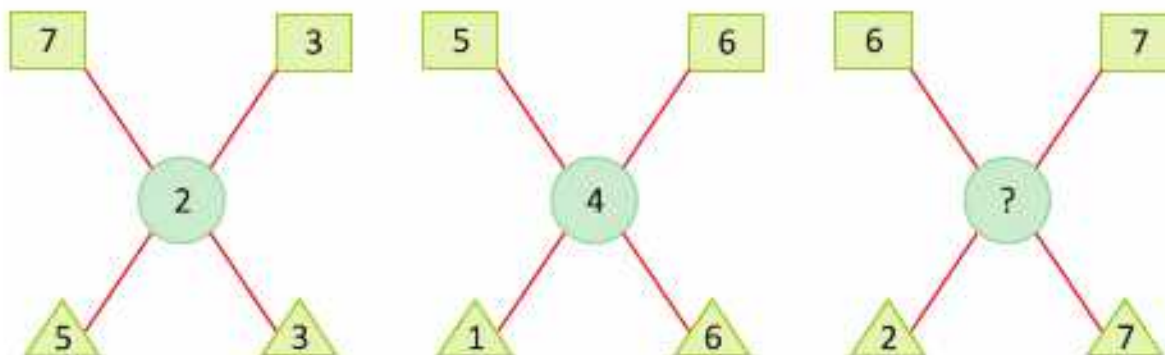
4. Change the Roman numeral IX into an even number by drawing just one letter.

5. What number is missing in the centre square?

5	4	1
3		3
2	2	6

6. My double exceeds my half by 99. What am I?

7. What number is missing in the third center circle?



Ans.

8. A party of people went to a restaurant. All of them ordered exactly the same meal. Their combined bill was ₹ 440.11. How many people were there in the party?

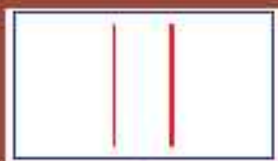


Odd One Out!

Odd refers to an thing which is not suitable or different at that place.

Tick (✓) the odd one out.

1



(a)



(b)



(c)



(d)

2



(a)



(b)



(c)

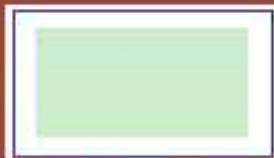


(d)

3



(a)



(b)



(c)



(d)

4



(a)



(b)



(c)



(d)



Mind Test

Mind test will tackle your knowledge.

A. Answer these puzzles.

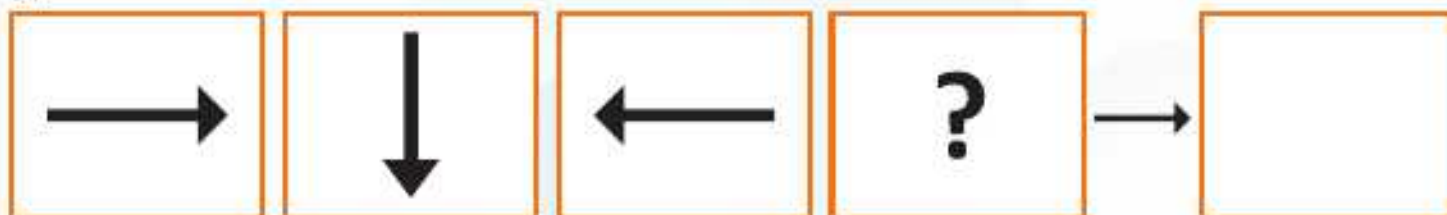
1. Vini's father has four daughters: Rini, Siya, Riya. What is the name of the fourth daughter?

2. How many scores make a century? _____
3. If 371 means 'Now' and 483 means 'PIN'. How would 'WIN' be written in numbers?

4. How many ₹ 3 stamps are there in a dozen? _____
5. Divide 30 by half and add 10. What do you get? _____
6. In a race, you overtake the third fastest person what position do you finish? _____

B. Complete the sequences below.

1. XXXXXOO, XXXXOOO, XXXOOOO? _____
2. 2, 6, 12, ...? _____
- 3.

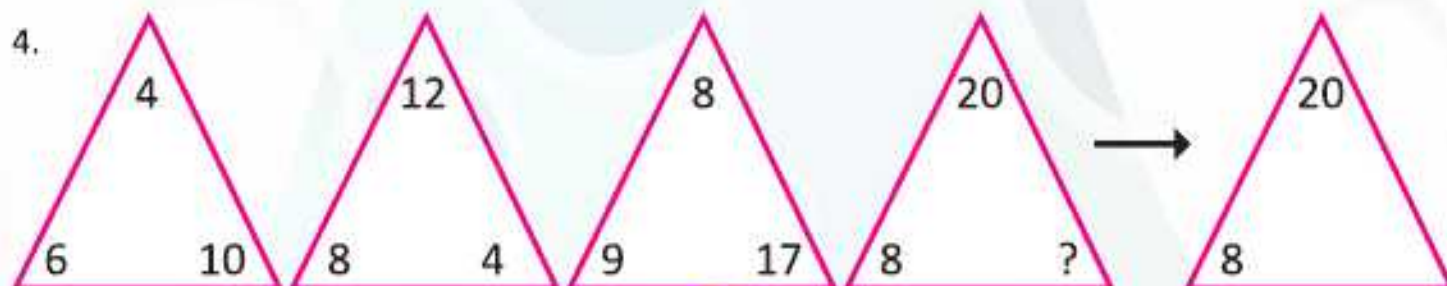


(a)

(b)

(c)

(d)





Letter Coding

In the questions given below, the letters in a word are replaced by certain other letters according to a specific rule to form its code. Detect the coding pattern rule and answer the questions accordingly.

1. In a certain code, TEACHER is written as VGCEJGT. How is CHILDREN written in that code?
(a) EJKNFTGP ☐ (b) EJKNFGTO ☐ (c) RJKNEGTP ☐
2. In a certain code, ROAD is written as URDG. How is SWAN written in that code?
(a) VXDQ ☐ (b) VZDQ ☐ (c) VZCP ☐
3. If HEALTH is written as GSKZDG, how will NORTH be written in that code?
(a) OPSUT ☐ (b) FRPML ☐ (c) GSQNM ☐
4. In a certain code language, BEAT is written as YUZG, what will be the code of MILD?
(a) ONRW ☐ (b) NOWR ☐ (c) NROW ☐
5. If FISH is written as EHRG in a certain code, how would JUNGLE be written in the code?
(a) ITMFKD ☐ (b) ITNFKD ☐ (c) TIMFKD ☐
6. In a certain code language, THANKS is written as SKNTHA. How is STUPID written in that code language?
(a) DIPUTS ☐ (b) DISPUT ☐ (c) DIPSTU ☐



Answer the following questions.

5. Read the clues and write the answer in the boxes provided. When you read the first. Column of each group of answers, you will see a special message.

- | | | | | | |
|----|--|--|--|--|------------------------|
| 11 | | | | | knitters need |
| 12 | | | | | a metal element |
| 13 | | | | | a baby sheep |
| 14 | | | | | canines |
| 15 | | | | | the king of the jungle |
| 16 | | | | | measurement |
| 17 | | | | | speedy |
| 18 | | | | | a great lake |

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Direct Analogy

Analogy is a comparison of two otherwise unlike things based on similarity of a particular aspect.

- Part is related to **whole** in the same way **Arc** is related to _____
 (a) Rectangle  (b) Circle  (c) Triangle 
- Tennis** is related to **court** in the same way **Boxing** is related to _____
 (a) Pool  (b) Ring  (c) Arena 
- Doctor** is related to **patient** in the same way **Lawyer** is related to _____
 (a) Customer  (b) Accused  (c) Client 
- Needle** is related to **thread** in the same way **Pen** is related to _____
 (a) Ink  (b) Cap  (c) Paper 
- Soap** is related to **wash** in the same way as **Broom** is related to _____
 (a) Clean  (b) Dust  (c) Sweep 
- Cat** is related to **kitten** in the same way **Owl** is related to _____
 (a) Fry  (b) Fawn  (c) Outlet 
- Thick** is related to **Thin** in the same way **Idle** is related to _____
 (a) Business  (b) Activity  (c) Industrious 
- Chef** is related to **Restaurant** in the same way is related to **Druggist**.
 (a) Medicine  (b) Pharmacy  (c) Chemist 

9. **Hare** is related to **Burrow** in the same way **owl** is related to _____
- (a) Barn (b) Hive (c) Nest
10. **Umbrella** is related to **rain** in the same way **Goggles** is related to _____
- (a) Glare (b) Stare (c) Sight
11. **Mathematics** is related to **numbers** in the same way **History** is related to _____
- (a) Events (b) Dates (c) Wars
12. **Afghanistan** is related to **Kabul** in the same way as **Australia** is related to _____
- (a) Airana (b) Tirana (c) Vienna
13. **Iron** is related to **solid** in the same ways as **Mercury** is related to _____
- (a) Liquid (b) Solid (c) Gas
14. **Lead** is related to **Pencil** in the same way as **Ink** is related to _____
- (a) Pot (b) Pen (c) Stamp
15. **Bull** is related to **Draught** in the same way as **Cow** is related to _____
- (a) Milch (b) Farm (c) Lives tock





All Occupations

The work of a person does to earn a living is known as occupation.

What is the name given to the occupations of the people who do the following jobs? Answer in one word.

1. A person in charge of a prison.
2. A wine merchant who sells wine.
3. A male servant that attends a noble man or gentleman.
4. A person who makes drawings and plans.
5. A person who stitches or repairs clothes.
6. A bookseller, seller of paper and writing impalements.
7. A constructor or repairer of ship.
8. A person who removes the fleece from sleep.
9. A ship's officer in charge of provision and accounts.
10. A person who gathers information and write news stories.
11. A maker of ladies hats and bonnets etc.
12. A maker and dealer of gold articles.
13. A farm worker who looks after cows.
14. A sweet and candy maker.
15. A person who decorates books with gold leaf.

Milliner
Warden
Byreman
Reporter

Puerser
Valet
Goldsmith
Stationer

Book Glider
Vinter
Shearman
Shipwright

Drafter
Confectioner
Tailor

HELP BOX



Proverbs

Proverbs are short clever sentences that usually offer life advice.

Write the meaning of these following proverbs.

1. Actions speak louder than words.

2. All good things must come to an end.

3. A picture is worth a thousand words.

4. Birds of a feather flock together.

5. Don't judge a book by its cover.

6. Good things come to those who wait.

7. There is no time like the present.

8. You can't always get what you want.

Idioms

An Idiom is a common word or phrase which means something different from its literal meaning but can be used understood because of their popular use.

Match the idiom with its real like meaning.

- | | |
|--|---|
| 1. _____ Break a leg | (a) I'm jealous |
| 2. _____ Don't cut corners | (b) Good luck |
| 3. _____ My words went in one ear and
out the other | (c) It's raining hard |
| 4. _____ Hang in these | (d) You're joking |
| 5. _____ I'm green with envy | (e) Do your best instead of
skipping something |
| 6. _____ Time flies | (f) You make me laugh |
| 7. _____ Don't spill the beans | (g) Don't talk |
| 8. _____ It's raining cats and dogs | (h) You didn't listen to me |
| 9. _____ I'm under the weather | (i) Don't give up |
| 10. _____ Hold your tongue | (j) I'm sick |
| 11. _____ You're pulling my leg | (k) Times goes quickly |
| 12. _____ you crack me up | (l) Don't tell me secret |



Literary Terms

Literary terms refer to the technique, style and formatting used by writers and speakers to masterfully emphasize, embellish or strengthen their compositions.

Identify these literary terms and give the correct answer with the help of clues.

1. The method an author uses to acquaint the reader with his characters
_____.
2. A conversation between characters in a story, novel, play, poem or work of nonfiction.
_____.
3. A stated comparison between two things that are really very different but share some common element. _____
4. A series of stressed and unstressed sound in a group of words. It may be regular or may be varied. _____
5. Any language which deviates from literal language so as to furnish novel effects or fresh insights into the subject being discussed. _____
6. Repeated consonant sounds occurring at the beginning of words or within words.

7. A form of metaphor in which a lifeless objects, an animal or an idea is made to act like a person. _____

Alliteration

Simile

Personification

Characterization

Rhythm

Figurative Language

Dialogue

HELP BOX

Homophones

A homophone is a word that is pronounced the same as another word but has a different meaning.

Tick (✓) the homophones that best fits the sentence.

1. My parents (allowed / aloud) me to watch a movie with my friend.
2. I had (eight / ate) dollars left to play video games.
3. My favorite team had (one / won) the world series.
4. We stayed at (inn/in) overnight.
5. My mom waited for the (sale / sail) at the store.
6. I wanted to go to the movies at (knight / night).
7. I (led / lead) my friend to the bench where I liked to eat lunch.
8. The (sun / son) rose in the morning.
9. (Some / Sum) of the things I like to do the most are play video games, watch TV, and do my homework.
10. Our team had (to / two) more games left in the season.





Words Often Confused

Many words in English often Confused the children because they sound same, but have different meanings.

Fill the correct words in the blank spaces.

1. Accede : ; exceed : surpass.
2. Accept : to receive; ; other than.
3. Coarse : ; course : syllabus
4. Principal : chief of the College; : law of rule.
5. Refuge : ; refuse : not to accept.
6. Route : road or path; : origin.
7. Stationary : ; stationery : writing materials.
8. Straight : direct; strait : .
9. Practice : the action of doing something; : to do again and again.
10. Maid : ; made : to make anything.
11. Naughty : ; knotty : difficult to solve.
12. Precede : to go before: : to continue.

Root
Principle

Practise
Consent

Proceed
Shelter

Wicked
Rough

Female servant
Fixed

Except
Narrow

HELP BOX

ANSWER SHEET

Chapter-1 : 1. Linguistics; 2. toxicology; 3. Dermatology; 4. Botany; 5. Psychology; 6. Hydrology; 5. Sociology; 8. acoustics; 9. Chemistry; 10. Zoology; 11. Geology; 12. Ornithology; 13. Biology; 14. Entomology; 15. Astrology; **Chapter-2 :** 1. (b); 2. (a); 3. (c); 4. (c); 5. (a); **Chapter-3 :** 1. (a); 2. (c); 3. (a); 4. (c); 5. (b); **Chapter-4 :** 1. Alligator snapping turtle; 2. Electric Ray; 3. Green Mamba; 4. Angles Fish; 5. Antlion; **Chapter-5 :** 1. Swordfish; 2. Clownfish; 3. Surgeonfish; 4. Parrotfish; **Chapter-6 :** 1. Earth worm; 2. beak; 3. African Elephant; 4. Humpback; 5. 2 year; 6. no; 7. 20 Times; 8. Freshwater; 9. 20 hours; 10. one-third; **Chapter-7 :** 1. Poodle; 2. Doberman; 3. Bullmastiff; 4. Dalmatian; 5. Blue Heeler; 6. Maltese; 7. Chihuahua; 8. Bulldog; 9. Pug; **Chapter-8 :** 1. Cattle; egret; 2. Peacock; 3. spot-billed Duck; 4. Koel; 5. Sparrow; **Chapter-9 :** 1. Ostrich; 2. colugo; 3. Woolly Flying squirrel; 4. Draco Volans; 5. Bats; **Chapter-10 :** 1. Pencil (a) crayon (b) 2. Collage; 3. Water; Oil colours; 4. Weaving; 5. Stamping; **Chapter-11 :** 1. Ceramics; 2. Graffiti; 3. Animation; 4. Photography; 5. Sculpture 6. Architecture; **Chapter-12 :** 1. Common Operating Machine Purposely used for tech Technological and Educational Research; 2. Central Processing Unit; 3. Programmable Read only Memory; 4. Hard Disk Drive; 5. Switched Mode Power Supply; 6. Basic Input Output system; 7. Solid State Drive; **Chapter-13 :** 1. Nausadar 2. Gypsum; 3. Vinegar; 4. Washing soda; 5. Laughing Gas; 6. Chalk 7. Sugar; 8. Chloroform; 9. Bleaching Powder; 10. Caustic Soda; 11. Ethene; 12. Alcohol; 13. Sand; 14. TNT; 15. Salt; **Chapter-14 :** 1. Rocket; engine; 2. Comet; 3. Astronomy; 4. Spaceshuttle; 5. Crator; 6. Nabula; 7. Orbit; 8. Satellite; **Chapter-15 :** 1. 19 April, 1975; 2. ISRO; 3. Bangaluru; 4. 91; 5. batteries; 6. Downlink; 7. 256; 8. 96.46; 9. commication; 10. Internal; 11. 11 February 1992; 12. Kosmas; **Chapter-16 :** 1. Robert Koffler Jarvik; 2. J Robert Oppenheimer; 3. Lazzlo Biro; 4. Rudolf Diesel; 5. Henry Seely white; 6. Igor Sikorsky 7. Edward W. Kellogg and Chester W. Rice; 8. Nicolas-Jacques Conte; 9. Johannes Gutenberg; 10. King Camp Gillette; 11. James Watt; 12. Hubert Cecil Booth; 13. Johan Vaaler; 14. Ralph Baer; 15. Benjamin Franklin; 16. Wilhelm Kriss; 17. Hiram Maxim; 18. Alexander Ferdinand Godefroy; **Chapter-17 :** Do yourself; **Chapter-18 :** 1. (b); 2. (c); 3. (b); 4. (a); 5. (a); **Chapter-19 :** 1. Water Hyacinth; 2. Mistletoe; 3. Mangrove; 4. Resurrection Fern; 5. Saguaro; **Chapter-20 :** 1. Mumbai; 2. Bengaluru; 3. Jaipur; 4. Dhanbad; 5. Lucknow; 6. Ajmer; 7. Hyderabad; 8. Kolkata; **Chapter-21 :** 1. Tamil Nadu; 2. Goa; 3. Goa; 4. Sikkim; 5. Arunachal Pradesh; 6. Rajasthan; 7. Uttarakhand; 8. Laddakh; 9. Aasam; 10. Bihar; 11. Andhara Pradesh; 12. Odisha; 13. Punjab; **Chapter-22 :** 1. (b); 2. (a); 3. (a); 4. (a); 5. (b); 6. (a); 7. (a); 8. (a); 9. (a); 10. (b); 11. (a); **Chapter-23 :** 1. Indian Machiavelli; 2. Nightingale of India; 3. Flying Sikh; 4. Swar Kokila; 5. Netaji; 6. Haryana Hurricane; 7. Tuti-yi-Hind; 8. Missile Man; 9. Birdman of India; **Chapter-24 :** 1. Brahmaputra River; 2. Ganga River 3. Yamuna River; 4. Kaveri River; 5. Krishan River; **Chapter-25 :** Chanakya; 2. Charak; 3. Bhaskaracharya; 4. Panini; 5. Sushruta; **Chapter-26 :** Mumbai; 2. Bengaluru; 3. Jaipur; 4. Dhanbad; 5. Lucknow; 6. Ajmer; 7. Hyderabad; 8. Kolkata; **Chapter-27 :** 1. Rudrama Devi; 2. Rani Durgawati; 3. Chand Bibi; 4. Ahilya Bai Holkar; 5. Rani Chennamma; 6. Rani Laxmi Bai; 7. Rani Avanti Lodhi; **Chapter-28 :** A. 1. Darjeeling Himalayan Railway; 2. Tagore Hill; 3. Bodhi Gaya Temple; 4. Victoria Memorial; 5. Konark Sun Temple; 6. Jagannath temple; 7. Cellular Jail; 8. National Library of India; **Chapter-29 :** 1. Guru; 2. Guru Nanak Dev; 3. Mahaveer; 4. Easter; 5. Muhammad's first revelation from God; **Chapter-30 :** 1. Raj Ghat; 2. Sant Vani; 3. Vijay Ghat; 4. Veer Bhumi; 5. Karma Bhumi; **Chapter-31 :** 1. Aryabhatta; 2. Bhaskaracharya; 3. Srinivasa Ramanujan; **Chapter-32 :** 1. Lake Baikal; 2. Atacama Desert; 3. Statue of Unity; 4. Pacific Ocean; 5. Mammoth Cave, Kentucky; **Chapter-33 :** 1. Plato; 2. Copernicus; 3. Herodotus; 4. Archimedes; 5. Sophocles; **Chapter-34 :** 1. (b); 2. (b); 3. (c); 4. (a); 5. (b); **Chapter-35 :** 1. 12; 2. 21; 3. 156; 4. 7; 5. 3 : 20; **Chapter-36 :** 1. 89; 153; 121; 2. 1000; 3. 25; 4. Six; 5. 4; 6. 66; 7. 4; 8. 5; **Chapter-37 :** 1. (c); 2. (a); 3. (d); 4. (c); **Chapter-38 :** 1. Vini; 2. 100; 3. 183; 4. 36; 5. 70; 6. third; 8. 1. XX0000; 2. 72; 3. () 4. 12; **Chapter-39 :** 1. (a); 2. (b); 3. (c); 4. (c); 5. (c); 6. (c); **Chapter-40 :** 1. Mistaken; 2. (a), MISUNDERSTOOD, (b). GOOD AFTERNOON, (c). FOREIGN LANGUAGE, (d). TRICYCLE, (e). TOO BIG TO IGNORE; 3. Crystal; 4. Startling, stalling, sting, sting, sing, sin, in, 5. 1. POOL, 2. RICH, 3. OVER, 4. TRAP, 5. EXIT, 6. CANE, 7. TALL, 8. ONCE, 9. UNDO, 10. REAL, 11. WOOL, 12. IRON, 13. LAMB, 14. DOGS, 15. LION, 16. INCH, 17. FAST, 18. ERIE. **Message:** PROTECT OUR WILDLIFE; **Chapter-41 :** 1. (b); 2. (b); 3. (c); 4. (a); 5. (c); 6. (c); 7. (c); 8. (b); 9. (a); 10. (a); 11. (a); 12. (c); 13. (a); 14. (b); 15. (a); **Chapter-42 :** 1. Warden; 2. Vinter; 3. Valet; 4. Drafter; 5. Tailor; 6. Stationer; 7. Shipwright; 8. Shearman; 9. Purser; 10. Reporter; 11. Milliner; 12. Goldsmith; 13. Byreman; 14. Confectioner; 15. Book Glider; **Chapter-43 :** 1. What Someone does has more value than what someone says; 2. Nothing lasts forever, everything gets completed with a conclusion; 3. A- photo can tell you more than any word can; 4. Similarities are attractive; 5. one should not judge the value of something or someone by their outward appearance alone; 6. People who wait patiently are rewarded and often achieve their desires and goals; 7. Something should be done now rather than later; 8. sometimes you have to accept that you will not be able to have; **Chapter-44 :** 1. (b); 2. (e); 3. (h); 4. (l); 5. (a); 6. (k); 7. (l); 8. (c); 9. (j); 10. (g); 11. (d); 12. (f); **Chapter-45 :** 1. Characterization; 2. Dialogue; 3. Simile; 4. Rhythm; 5. Figurative Language; 6. Alliteration; 7. Personification; **Chapter-46 :** 1. Allowed; 2. eight; won; 4. inn; 5. sale; 6. might; 7. led; 8. sum; 9. some; 10. two; **Chapter-47 :** 1. Consent; 2. Except; 3. Rough; 4. Principle; 5. Shelter; 6. Root; 7. Fixed; 8. Narrow; 9. Practise; 10. Female servant; 11. Wicked; 12. Proceed